

4.4, 5.1–5.2 Orthonormal Bases, QR, and Determinants

Concise exam revision notes for orthonormal columns, QR factorization, and determinant rules.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Orthonormal columns make projection and least squares simple. Determinants measure whether a square matrix is singular and how elimination changes volume.

This note has two connected parts:

$$Q^T Q = I \quad \text{and} \quad \det A.$$

The first simplifies geometry; the second summarizes invertibility for square matrices.

2 Orthogonal vs Orthonormal

Word	Condition	Meaning
Orthogonal	$q_i^T q_j = 0$ for $i \neq j$.	Different directions are perpendicular.
Orthonormal	$q_i^T q_j = 0$ and $q_i^T q_i = 1$.	Perpendicular and unit length.
Matrix Q with orthonormal columns	$Q^T Q = I$.	Dot products are already built in.

If Q is square and orthonormal, then

$$Q^{-1} = Q^T.$$

For rectangular Q , $Q^T Q = I$, but $Q Q^T$ is a projection matrix.

3 Gram–Schmidt

Gram–Schmidt turns independent vectors into orthogonal vectors, then normalizes them.

For two vectors a, b , first keep

$$A = a.$$

Remove from b the part in the a -direction:

$$B = b - \frac{a^T b}{a^T a} a.$$

Then normalize:

$$q_1 = \frac{A}{\|A\|}, \quad q_2 = \frac{B}{\|B\|}.$$

For a third vector c , subtract the projections onto the previous orthogonal directions:

$$C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B.$$

Exam trigger. If the problem says “make an orthonormal basis”, use Gram–Schmidt, then divide by lengths.

4 QR Factorization

For independent columns,

$$A = QR,$$

where Q has orthonormal columns and R is upper triangular.

The entries of R record dot products with the q ’s:

$$R = Q^T A.$$

For least squares,

$$A^T A \hat{x} = A^T b$$

becomes

$$R \hat{x} = Q^T b.$$

This is easier because R is triangular.

5 Projection with Orthonormal Columns

If Q has orthonormal columns, projection onto $C(Q)$ is

$$p = QQ^T b.$$

In vector form,

$$p = (q_1^T b)q_1 + (q_2^T b)q_2 + \cdots + (q_n^T b)q_n.$$

Column type	Projection habit
Orthonormal q_i	Use $q_i^T b$ directly.
Orthogonal but not unit	Divide by $a_i^T a_i$.
General independent columns	Use $A(A^T A)^{-1} A^T$.

6 Orthogonal Matrices

For square orthogonal Q ,

$$Q^T Q = I, \quad Q^{-1} = Q^T.$$

Orthogonal matrices preserve lengths and dot products:

$$\|Qx\| = \|x\|, \quad (Qx)^T (Qy) = x^T y.$$

Rotations and reflections are the main exam examples.

7 Determinant Rules

Operation or fact	Effect on determinant
Exchange two rows	Changes the sign.
Multiply one row by c	Multiplies determinant by c .
Add a multiple of one row to another	No change.
Triangular matrix	Product of diagonal entries.
Singular matrix	Determinant is 0.
Product	$\det(AB) = \det A \det B$.
Transpose	$\det(A^T) = \det A$.

Exam memory. A zero determinant means no inverse. A nonzero determinant means the square matrix is invertible.

8 Pivots and Cofactors

If elimination uses no row exchanges, then

$$\det A = (\text{product of pivots}).$$

For leading submatrices A_k , pivots can be checked by determinant ratios:

$$d_k = \frac{\det A_k}{\det A_{k-1}}.$$

Cofactor expansion is useful when a row or column has many zeros:

$$\det A = \sum_j (-1)^{i+j} a_{ij} \det M_{ij}.$$

Use it selectively; elimination is usually faster for larger matrices.

9 Common Mistakes

Mistake	Correct exam habit
Calling orthogonal columns orthonormal.	Check both perpendicularity and unit length.
Using $Q^{-1} = Q^T$ for rectangular Q .	Only square orthogonal matrices have inverse Q^T .
Forgetting to normalize after Gram–Schmidt.	Orthogonal first, unit length second.
Using QQ^T for any basis.	Use it only for orthonormal columns.
Changing a row and forgetting the determinant effect.	Track swaps, scaling, and row replacement separately.
Expanding cofactors when elimination is simpler.	Use zeros or structure to choose the method.

10 Final Exam Checklist

Checklist.

1. Check whether vectors are orthogonal by dot products.
2. Check whether vectors are orthonormal by dot products and lengths.
3. Use Gram–Schmidt to build orthogonal vectors.
4. Normalize each vector to get q_i .
5. Form Q from the q_i 's and compute $R = Q^T A$.
6. For least squares with $A = QR$, solve $R\hat{x} = Q^T b$.
7. For determinants, choose elimination unless cofactor expansion has many zeros.
8. Track determinant changes from row operations.
9. Use $\det A = 0$ to identify singular square matrices.